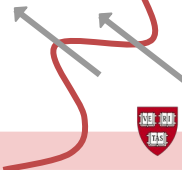


Counterfactual Explanations Without Opening the Black Box

Paper by: Sandra Wachter, Brent Mittelstadt, & Chris Russell
Presented by: Alex, Chelse, & Usha



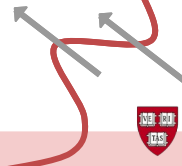
Introduction + Motivation

- EU General Data Protection Regulation (2018)
 - “the toughest privacy and security law in the world”
 - Article 13-14, regarding automated decision-making: “meaningful information about the logic involved”
 - Recital 71: “the right to obtain an explanation of the decision reached and to challenge the decision”



4 key problems

1. Not legally binding
2. Only applicable in limited cases
3. Explainability is technically very challenging
4. Competing interests of data controllers, subjects, etc

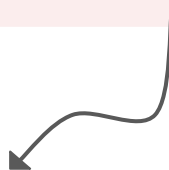


Unconditional Counterfactual Explanations

Authors propose

- 3 aims for explanations
 - Inform and help the subject understand “why”
 - Provide grounds to contest decisions
 - Understand options for recourse
- CFEs
 - Fulfill the above goals
 - Overcome challenges w.r.t. current interpretability work
 - Can bridge the gap between interests of data subjects and data controllers

Do you buy
this?

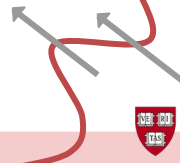


Unconditional Counterfactual Explanations

- Definition:
 - A statement of how the world would have to be different for a desirable outcome to occur
 - **Because** of features $\{x_1, x_2\}$, x 's outcome was label y_1
 - **If** $x = \{x_1 + \delta_1, x_2 + \delta_2\}$, **then** y_2 .
- Example:
 - “You were denied a loan because your annual income was \$30000. If your income has been \$45000, you would have been offered a loan”

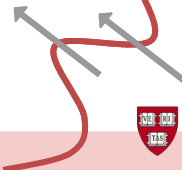
Notes:

- No one **unique** CF
- Need to consider **actionability** (mutability of variables)
- Providing **several diverse** CFEs may be more useful than simply the closest/shortest one



Background + Related Work

- Historic context of knowledge
- Prior explainability work
- Adversarial examples/perturbations
- Causality/Fairness

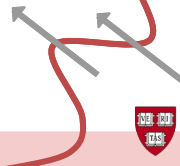


Background: Historic Context of Knowledge

- In order to know something, it is not enough to simply **believe** that it is true: rather, you must also have a good **reason** for believing it
- If q were false, S would not believe p

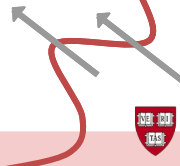
Note:

- This statement only describes S 's **beliefs**, which might not reflect reality
- This statement can be made without knowledge of the **causal relationship** between p and q
- Q: who is S ? The user? The model? Us?



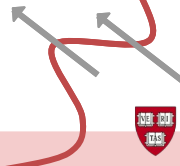
Background: Previous Explanations in AI/ML

- Previously: providing insight into the internal state of an algorithm, human-understandable approximations of the algorithm
- Three-way tradeoff between
 1. quality of approximation
 2. ease of understanding the function
 3. size of the domain for which the approximation is valid
- CFEs:
 - Minimal amount of information
 - Require no understanding of internal logic of model
 - No approximation (although might not always be minimum length)
 - Con: May be overly restrictive



Background: Adversarial Perturbations

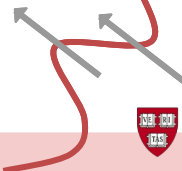
- Small changes to a given image can result in the image being assigned to a different class
- Very similar to CFEs but the changes aren't necessarily sparse
- Often not human perceptible
 - Authors propose that this is because the new images lie outside the “real-image” manifold
 - Emphasize the importance of solutions/CFEs being *possible* as well as close
 - Further research into structure of high-D data is structured before CFEs can be useful/reliable in those domains



Background: Causality and Fairness

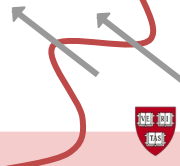
- Can provide evidence that models/decisions are affected by protected attributes
- If CFEs change one's race, the treatment of that individual is dependent on race

Is the
converse
true?



Summary of Contributions

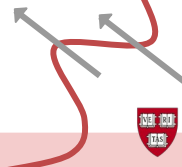
- Highlights the difficulties with conveying the inner workings of modern ML algorithms to users
 - Complexity
 - Lack of utility (except for “builders”)
- Introduces an algorithmic approach to counterfactuals
 - Rooted in adversarial machine learning
- Connects counterfactuals to the GDPR
 - Demonstrates advantages over other interpretability methods from a policy perspective



Approach: Counterfactual Optimization

$$\arg \min_{x'} \max_{\lambda} \lambda (f_w(x') - y')^2 + d(x_i, x')$$

- Minimize:
 - Squared error with desired (counterfactual) label
 - Distance to perturbed point
- While maximizing weight on squared error term
 - We want to have the prediction change more than we want the point to be close
 - Iteratively: solve for x' , then maximize λ
- Settings:
 - Tabular Data
 - Regression



Approach: Distance Metrics

$$d(x, x') = \sum_{k \in F} \frac{\|x_{i,k} - x'_k\|_p}{N}$$

Norms

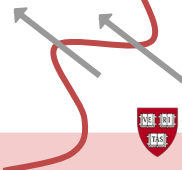
$$|x_{i,k} - x'_k|$$

Normalizing Factors

$$MAD_k := \text{median}_{j \in P} (|X_{j,k} - \text{median}_{l \in P}(X_{l,k})|)$$

$$(x_{i,k} - x'_k)^2$$

$$\text{std}_{j \in P}(x_{j,k})$$



Experimental Results: LSAT

$$d(x_i, x') \propto \sum_{k \in F} (x_{i,k} - x'_k)^2$$

Original Data				Unnormalised L2 Counterfactuals			Counterfactual Hybrid		
score	GPA	LSAT	Race	GPA	LSAT	Race	GPA	LSAT	Race
0.17	3.1	39.0	0	3.0	39.0	0.3	1.5	38.4	0
0.54	3.7	48.0	0	3.5	47.9	0.9	-1.6	45.9	0
-0.77	3.3	28.0	1	3.5	28.1	-0.3	5.3	28.9	0
-0.83	2.4	28.5	1	2.6	28.6	-0.4	4.8	29.4	0
-0.57	2.7	18.3	0	2.9	18.4	-1.0	8.4	20.6	0

Table 1 - Unnormalized L2

				Normalised L2					
score	GPA	LSAT	Race	GPA	LSAT	Race	GPA	LSAT	Race
0.17	3.1	39.0	0	3.0	37.0	0.2	3.0	34.0	0
0.54	3.7	48.0	0	3.5	39.5	0.4	3.5	33.1	0
-0.77	3.3	28.0	1	3.5	39.8	0.4	3.4	33.4	0
-0.83	2.4	28.5	1	2.7	37.4	0.2	2.6	35.7	0
-0.57	2.7	18.3	0	2.8	28.1	-0.4	2.9	34.1	0

Table 2 - Normalised L2

Experimental Results: LSAT

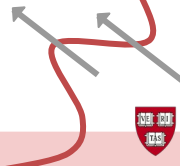
$$d(x_i, x') = \sum_{k \in F} \frac{|x_{i,k} - x'_k|}{MAD_k}$$

score	Original Data			Normalised L1			Counterfactual Hybrid		
	GPA	LSAT	Race	GPA	LSAT	Race	GPA	LSAT	Race
0.17	3.1	39.0	0	3.1	35.0	0.1	3.1	34.0	0
0.54	3.7	48.0	0	3.7	33.5	0.0	3.7	32.4	0
-0.77	3.3	28.0	1	3.3	34.4	0.1	3.3	33.5	0
-0.83	2.4	28.5	1	2.4	39.3	0.2	2.4	35.8	0
-0.57	2.7	18.3	0	2.7	35.8	0.1	2.7	34.9	0

Table 3 - Normalised L1

Experimental Results: Discussion

- Realistic (or achievable) counterfactual values
 - Clipping and Clamping
- Counterfactual Targets
 - LSAT: Avg. Grade 0
 - Students would want to know how to improve their average grade
 - PIMA: Risk Score of 0.5
 - Patients would want to know how to achieve a lower risk score



Explanations and The GDPR

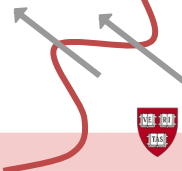
GDPR Requirements: Recital 71

- Implement suitable safeguards against automated decision-making
- Include specific information to the data subject and the right to obtain human intervention
 - To express their point of view
 - To challenge the decision
 - To obtain an *explanation* of the decision reached after such assessment

- Does not require opening the “black box” to explain the internal logic of decision-making systems
- Does not explicitly define requirements for explanations of automated decision-making

Counterfactual Explanations and The GDPR

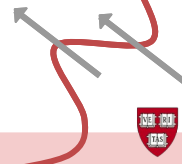
- Legislators wanted to clarify that some type of explanation can voluntarily be offered after a decision has been made
- Many aims for explanations are feasible
- Emphasis of GDPR is on protections and rights for individuals



Advantages of Counterfactual Explanations

- Bypass explaining the *internal workings* of complex machine learning system
- Simple to compute and convey
- Provide information that is both easily digestible and practically useful
 - Understanding the reasons for a decision
 - Contesting decisions
 - Altering future behaviour to receive a preferred outcome

Possible mechanism to meet the explicit requirements and background aims of the
GDPR



Broader Possibilities with the Right of Access

Art 15.

- Confirm whether or not personal data used
- Provide information available after a decision has been made
- Avoid disclosing personal data of other data subjects
- Balance interest of the subject and controller
 - Potential to contravene trade secrets or intellectual property rights



Broader Possibilities with the Right of Access

Counterfactuals

- Data of other data subjects or detailed information about the algorithm does not need to be disclosed
- Disclose only the influence of select external facts and variables on a specific decision
- Less likely to infringe on trade secrets or privacy

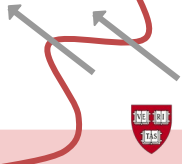


Understanding Through Counterfactuals

Art 12(1)

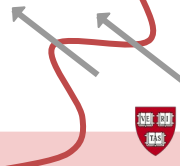
- Requires information to be conveyed in a “concise, transparent, intelligible and easily accessible form”
- CFEs align with this requirement by providing simple “if-then” statements
- CFEs provide greater insight into the data subject’s personal situation as opposed to an overview tailored to a general audience

Minimally burdensome and disruptive technique to understand the rationale of specific decisions



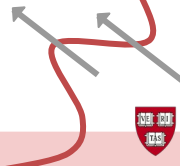
Legal Information Gaps on Contesting Decisions

- Art. 16
 - Data subject has the right to correct inaccurate data used to make a decision, but does not need to be informed of which data the decision depended
- Art. 22
 - Data subjects do not need to be informed of their right *not* to be subject to an automated decision



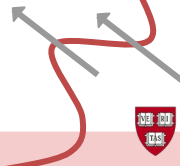
Contesting Through Counterfactuals

- Lead to greater protection for the data subject than currently envisioned by the GDPR
- Align with Article 29 Working Party
 - Understanding decisions + knowing legal basis is essential for contesting decisions
- Reduce burden on data subject
 - Understand most influential data (instead of vetting all collected data)
 - Compact way to convey dependencies



Explanations to Alter Future Decisions

- GDPR
 - Explanations not explicitly mentioned as a guide to altering behavior to receive a desired automated decision
- Article 29 Working Party
 - Provide suggestions on how to improve habits and receive better outcome
- Counterfactual explanations
 - Can address the impact of changes to more than one variable on a model's output at the same time
 - Can be used in a contractual agreement between data controllers and data subjects



Discussion

- Who do these explanations serve?
 - Developers, Data subjects, and Lawmakers
- Do you think they are misinterpreted?
 - Causality
 - Optimizing for sparsity may not reflect reality
- Priors and desired outcomes

“If it didn’t happen, why would I change my decision?”: How Judges Respond to Counterfactual Explanations for the Public Safety Assessment

Yaniv Yacoby,¹ Ben Green,² Christopher L. Griffin, Jr.,³ Finale Doshi-Velez¹

¹ Harvard University

² University of Michigan

³ James E. Rogers College of Law, University of Arizona

yanivyacoby@g.harvard.edu, bzgreen@umich.edu, chrisgriffin@arizona.edu, finale@seas.harvard.edu

Abstract

Many researchers and policymakers have expressed excitement about algorithmic explanations enabling more fair and responsible decision-making. However, recent experimental studies have found that explanations do not always improve human use of algorithmic advice. In this study, we shed light on how people interpret and respond to counterfactual explanations (CFEs)—explanations that show how a model’s output would change with marginal changes to its input(s)—in the context of pretrial risk assessment instruments (PRAIs). We ran think-aloud trials with eight sitting U.S. state court judges, providing them with recommendations from a PRAI that includes CFEs. We found that the CFEs did not alter the judges’ decisions. At first, judges misinterpreted the counterfactuals as real—rather than hypothetical—changes to defendants. Once judges understood what the counterfactuals meant, they ignored them, stating their role is only to make decisions regarding the actual defendant in question. The judges also expressed a mix of reasons for ignoring or following the advice of the PRAI without CFEs. These results add to the literature detailing the unexpected ways in which people respond to algorithms and explanations. They also highlight new challenges associated with improving human-algorithm collaborations through explanations.

Alongside this excitement about XAI, however, recent work has demonstrated that these benefits are not always realized in practice. User studies in multiple contexts have found that explanations did not help people evaluate the quality of algorithmic advice or incorporate that advice into decisions (e.g., Lai and Tan (2019); Bansal et al. (2021); Jacobs et al. (2021); Green and Chen (2019b)). Thus, it is not yet clear whether and under what conditions explanations improve human-algorithm decision-making. Moreover, most prior user studies have been limited to laypeople. It is particularly important to understand how practitioners interact with algorithmic explanations.

In this study, we investigated how sitting U.S. state court judges interact with a type of explanation known as “counterfactual explanations” (CFEs). CFEs provide a human decision-maker with information about how a model’s output changes based on variations in the input(s) (e.g., Martens and Provost (2014); Wachter, Mittelstadt, and Russell (2018); Goyal et al. (2019); Verma, Dickerson, and Hines (2020); Stepin et al. (2021)). In theory, this information could provide human decision-makers with a better understanding of how sensitive—or robust—the model is to marginal changes to its input(s). This insight could then



Algorithmic Recourse: from Counterfactual Explanations to Interventions

By Kamiri, Schölkopf, and Valera

Presentation by
Catherine Huang, Christina Xiao, Kelly Zhang

Algorithmic Recourse

- Increasingly algorithms are used to make consequential decisions for individuals
- Recourse: “*Systematic process of reversing unfavorable decisions made by algorithms and bureaucracies*”
 - Promoting **agency** and **trust**

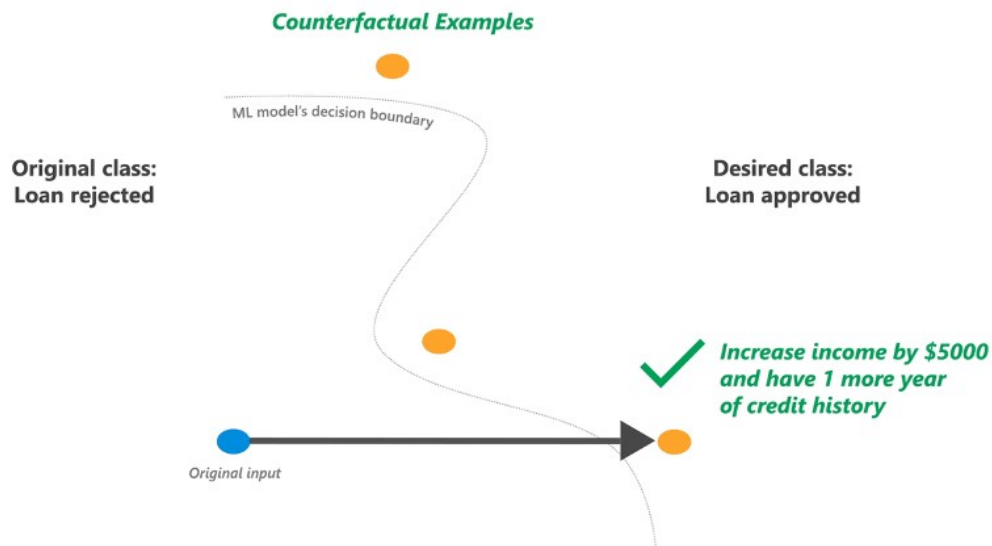


Paper's Main Contributions

- 1. Insufficiencies of previous problem formulations**
→ Motivates causal approach to recourse (Kelly)
- 2. Structural Causal Model approach to recourse (Catherine)**
- 3. Discuss examples motivated by real-world problems and future directions (Christina)**

Nearest Counterfactual Explanations

- For a person with features $\mathbf{x}^F$ who was denied a loan, find the “nearest neighbor” $\mathbf{x}$ who was granted a loan
- Difference between $\mathbf{x}^F$ and $\mathbf{x}$ is an “explanation” for the loan denial for $\mathbf{x}^F$



Nearest Counterfactual Explanation (CFE)

$$\mathbf{x}^{*CFE} \in \underset{\mathbf{x}}{argmin} \quad \text{dist}(\mathbf{x}, \mathbf{x}^F) \quad \text{s.t.} \quad h(\mathbf{x}) \neq h(\mathbf{x}^F), \mathbf{x} \in \mathcal{P},$$

- **Finding nearest counterfactual explanation as an optimization problem**
 - $\mathbf{x}^{*CFE}$ is the Counter Factual Explanation
 - Function $\mathbf{h}$ is the classifier
- **Distance metrics**
 - Lp norm; L1 norm divided by median absolute deviation; etc

Why are counterfactual **explanations** not ideal for **recommendations** for recourse?

Counterfactual Explanations provide **understanding** but do not necessarily lead to optimal **action recommendations**

Why are counterfactual **explanations** not ideal for **recommendations** for recourse?

1. Don't account for person's difficulty or "cost" of changing dimensions of $\mathbf{x}^F$ (addressed by Ustun et al.)
2. Don't account for downstream "causal" impact of taking actions (addressed by this work)

Accounting for the “Cost” of Actions (Ustun et al.)

$$\delta^* \in \underset{\delta}{\operatorname{argmin}} \operatorname{cost}(\delta; \mathbf{x}^F) \quad \text{s.t. } h(\mathbf{x}^{\text{CFE}}) \neq h(\mathbf{x}^F),$$

$$\mathbf{x}^{\text{CFE}} = \mathbf{x}^F + \delta,$$

$$\mathbf{x}^{\text{CFE}} \in \mathcal{P}, \quad \delta \in \mathcal{F},$$

- δ^* restricted to the set of “feasible changes”
- Consider linear impact of changes δ
- Non-trivial to choose costs that reflect people's' true objective functions

Insufficiencies of Ustun et al. Formulation

$$\delta^* \in \underset{\delta}{\operatorname{argmin}} \operatorname{cost}(\delta; \mathbf{x}^F) \quad \text{s.t. } h(\mathbf{x}^{\text{CFE}}) \neq h(\mathbf{x}^F),$$

$$\mathbf{x}^{\text{CFE}} = \mathbf{x}^F + \delta,$$

$$\mathbf{x}^{\text{CFE}} \in \mathcal{P}, \delta \in \mathcal{F},$$

1. Marginal cost of changing a feature is constant

- **Example:** Cost of going from salary \$0 to \$100k equals cost to go from salary \$800k to \$900k

Insufficiencies of Ustun et al. Formulation

2. Doesn't consider downstream “causal” impact of actions

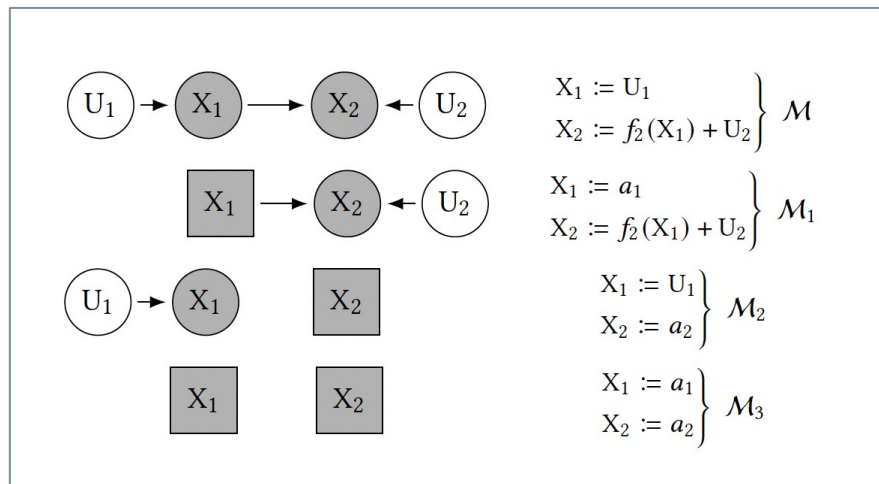
- **Example:** Loan decision for individual changed if “salary” reaches 100k (+33%) or “bank balance” reaches 30k (+20%).
20% change seems easier.
- However, best **action recommendation** is to increase salary by 14% when 30% of salary automatically saved to bank.

Actions as Interventions: Structured Causal Model (SCM)

Setup

- $M (M \in \Pi) = \langle F, X, U \rangle$: SCM
- X : endogenous (observed) variables
- U : exogenous (unobserved) variables
- $F: U \rightarrow X$, structural equations
- $A (\Pi \rightarrow \Pi)$: structural interventions, i.e. transformations between SCMs
 - of the form $A := \text{do}(\{X_i := a_i\}_{i \in I})$

Structural Interventions



- M : true world model
- observation: $M_1 \neq M_2 \neq M_3$

Actions as Interventions: Structural Counterfactuals

What will individual x^F 's feature vector be after the individual performs action set A in world M ?

Assumptions: (1) no hidden confounders (true SCM), (2) full access to invertible F

Idea: once we know F , X (endogenous variables) can be uniquely determined given U (exogenous variables)

Compute $F^{-1}(x^F)$

Takeaway: we can compute any structural counterfactual query for individual x^F :

$$x^{\text{SCF}} = F_A(F^{-1}(x^F)).$$

Limitations of CFE-Based Recourse: Formalism

Setup: x^F (individual features), δ^* action recommendation (Ustun et al. solution), I (set of indices of acted-upon observed variables: $I = \{i \mid \delta_i^* \neq 0\}$)

Definition (CFE-Based Actions): a set of structural interventions $A^{CFE} := \text{do}(\{X_i := x_i^F + \delta_i^*\}_{i \in I})$

gives
rise to

Proposition: $A^{CFE} \xrightarrow{\text{gives rise to}} x^{SCF} = x^{*CFE} := x^F + \delta^*$ (i.e. recourse is guaranteed) **if and only if I 's descendants = $\emptyset$.**

Corollary: if the true world M is independent—if all the observed features are root-nodes of G —then CFE-based actions always guarantee recourse.

Algorithmic Recourse via Minimal Interventions: Setup

Formulation

$$\begin{aligned} \mathbf{A}^* &\in \underset{\mathbf{A}}{\operatorname{argmin}} \quad \operatorname{cost}(\mathbf{A}; \mathbf{x}^F) \\ \text{s.t.} \quad &h(\mathbf{x}^{\text{SCF}}) \neq h(\mathbf{x}^F) \\ &\mathbf{x}^{\text{SCF}} = \mathbb{F}_{\mathbf{A}}(\mathbb{F}^{-1}(\mathbf{x}^F)) \\ &\mathbf{x}^{\text{SCF}} \in \mathcal{P}, \quad \mathbf{A} \in \mathcal{F}, \end{aligned}$$

Remarks

- ~~finding minimal shift of features~~ $\rightarrow$ finding minimal cost action set that yields favorable label
- $\mathbf{A}^* \in \mathcal{F}$ = set of feasible actions with minimally costly recourse
- $\operatorname{cost}(\square; \mathbf{x}^F): \mathcal{F} \times X \rightarrow \mathbb{R}_+$
- $\square^{\text{SCF}}: \mathbb{F}_{\mathbf{A}^*}(\mathbb{F}^{-1}(\square^F))$: resulting counterfactual
- $\square^{\text{SCF}} \neq \square^{\text{CFE}}$ (from Ustun et al.)!

Algorithmic Recourse via Minimal Interventions: Formalism

(from the
last slide)

$$\begin{aligned} A^* \in \underset{A}{\operatorname{argmin}} \quad & \operatorname{cost}(A; \mathbf{x}^F) \\ \text{s.t.} \quad & h(\mathbf{x}^{\text{SCF}}) \neq h(\mathbf{x}^F) \\ & \mathbf{x}^{\text{SCF}} = \mathbb{F}_A(\mathbb{F}^{-1}(\mathbf{x}^F)) \\ & \mathbf{x}^{\text{SCF}} \in \mathcal{P}, \quad A \in \mathcal{F}, \end{aligned}$$

Proposition:

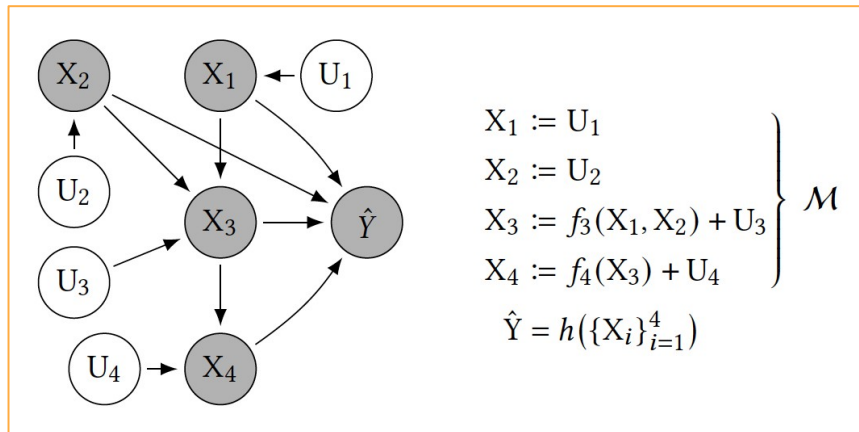
- A^{CFE} : Counter Factual Explanation-based action
- A^* : Minimal Intervention Solution (blue box above)

$$\operatorname{cost}(A^*; \mathbf{x}^F) \leq \operatorname{cost}(A^{\text{CFE}}; \mathbf{x}^F)$$

Algorithmic Recourse via Minimal Interventions: MINT

- Recourse through Minimal Interventions (MINT) idea:
 - Required: that we can compute structural counterfactual of an individual in the world given *any* feasible action
 - Focus on the case where the SCM is an additive noise model
 - $\Rightarrow$ Abduction-action-prediction technique (Pearl et al.) to compute x^{SCF} :
 $F_A(F^{-1}(x^F))$

Abduction-Action-Prediction to obtain x^{SCF}



$\{U_i\}_{i=1}^4$: mutually independent, exogenous

$\{f_i\}_{i=1}^4$: structural equations

$x = [x_1^F, x_2^F, x_3^F, x_4^F]^T$: observed factual features

(1) Abduction: compute exogenous variables

$$u_1 = x_1^F,$$

$$u_2 = x_2^F,$$

$$u_3 = x_3^F - f_3(x_1^F, x_2^F),$$

$$u_4 = x_4^F - f_4(x_3^F).$$

(2) Action: modify SCM with interventions

$$X_1 := [1 \in I] \cdot a_1 + [1 \notin I] \cdot U_1,$$

$$X_2 := [2 \in I] \cdot a_2 + [2 \notin I] \cdot U_2,$$

$$X_3 := [3 \in I] \cdot a_3 + [3 \notin I] \cdot (f_3(X_1, X_2) + U_3),$$

$$X_4 := [4 \in I] \cdot a_4 + [4 \notin I] \cdot (f_4(X_3) + U_4),$$

(3) Prediction: recursively compute

endogenous variables based on (1) and (2)

$$x_1^{\text{SCF}} := [1 \in I] \cdot a_1 + [1 \notin I] \cdot (u_1),$$

$$x_2^{\text{SCF}} := [2 \in I] \cdot a_2 + [2 \notin I] \cdot (u_2),$$

$$x_3^{\text{SCF}} := [3 \in I] \cdot a_3 + [3 \notin I] \cdot (f_3(x_1^{\text{SCF}}, x_2^{\text{SCF}}) + u_3),$$

$$x_4^{\text{SCF}} := [4 \in I] \cdot a_4 + [4 \notin I] \cdot (f_4(x_3^{\text{SCF}}) + u_4).$$

General Formulation and Solving the Optimization Problem

General Formulation

$$\mathbf{A}^* \in \underset{\mathbf{A}}{\operatorname{argmin}} \quad \operatorname{cost}(\mathbf{A}; \mathbf{x}^F)$$

$$\text{s.t.} \quad h(\mathbf{x}^{\text{SCF}}) \neq h(\mathbf{x}^F)$$

$$\begin{aligned} x_i^{\text{SCF}} = & [i \in I] \cdot (x_i^F + \delta_i) \\ & + [i \notin I] \cdot (x_i^F + f_i(\mathbf{pa}_i^{\text{SCF}}) - f_i(\mathbf{pa}_i^F)). \end{aligned}$$

$$\mathbf{x}^{\text{SCF}} \in \mathcal{P}, \quad \mathbf{A} \in \mathcal{F},$$

Remarks

- $x_i^F + \delta_i$: intervention
- $f_i(\mathbf{pa}_i^F)$: factual values of x_i 's parents
- $f_i(\mathbf{pa}_i^{\text{SCF}})$: counterfactual values of x_i 's parents
- new closed-form expression for $F_{\mathbf{A}^*}(F^{-1}(\mathbf{x}^F)) \Rightarrow$ use optimization methods

Experimental Setup

Synthetic setting:

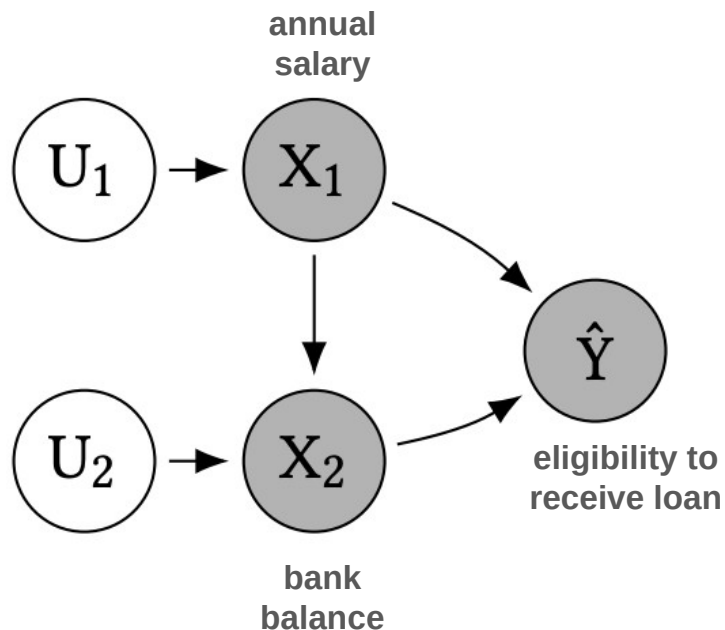
Generate data following
causal generative process

Real-world setting:

Use existing German credit
dataset to learn structural
causal model equations, by
fitting a linear regression

Cost for both is ℓ_1 norm over normalized feature change

Experimental Setup: Synthetic Setting



$$U_1 \sim \$10000 \cdot \text{Poisson}(10),$$

$$U_2 \sim \$2500 \cdot N(0,1)$$

$$\left. \begin{aligned} X_1 &:= U_1 \\ X_2 &:= f_2(X_1) + U_2 \end{aligned} \right\} \mathcal{M}$$

$$X_2 := 3/10 \cdot X_1 + U_2$$

$$\hat{Y} = h(X_1, X_2)$$

$$h = \text{sgn}(X_1 + 5 \cdot X_2 - \$225000)$$

Casual Generative Process

Experimental Results: Synthetic Setting

[annual salary, bank balance]

$\mathbf{x}^F$		$[\$75000, \$25000]^T$
Karimi et al.'s formulation	$\mathbf{A}^*$	$\text{do}(X_1 := x_1^F + \$10000)$
	$\mathbf{x}^{*\text{SCF}}$	$[\$85000, \$28000]^T$
Prior formulation	δ^*	$[\$0, +\$5000]^T$
	$\mathbf{x}^{*\text{CFE}}$	$[\$75000, \$30000]^T$

Experimental Results: Synthetic Setting

[annual salary, bank balance]

$\mathbf{x}^F$		[\$75000, \$25000] ^T
Karimi et al.'s formulation	$\mathbf{x}^{*SCF}$	[\$85000, \$28000] ^T
Prior formulation	$\mathbf{x}^{*CFE}$	[\$75000, \$30000] ^T

$\mathbf{x}^{*SCF}$ further dist from $\mathbf{x}^F$ than $\mathbf{x}^{*CFE}$

BUT

$$\text{cost}(\delta^*; \mathbf{x}^F) \approx 2 \text{ cost}(\mathbf{A}^*; \mathbf{x}^F)$$

Experimental Setup

Synthetic setting:

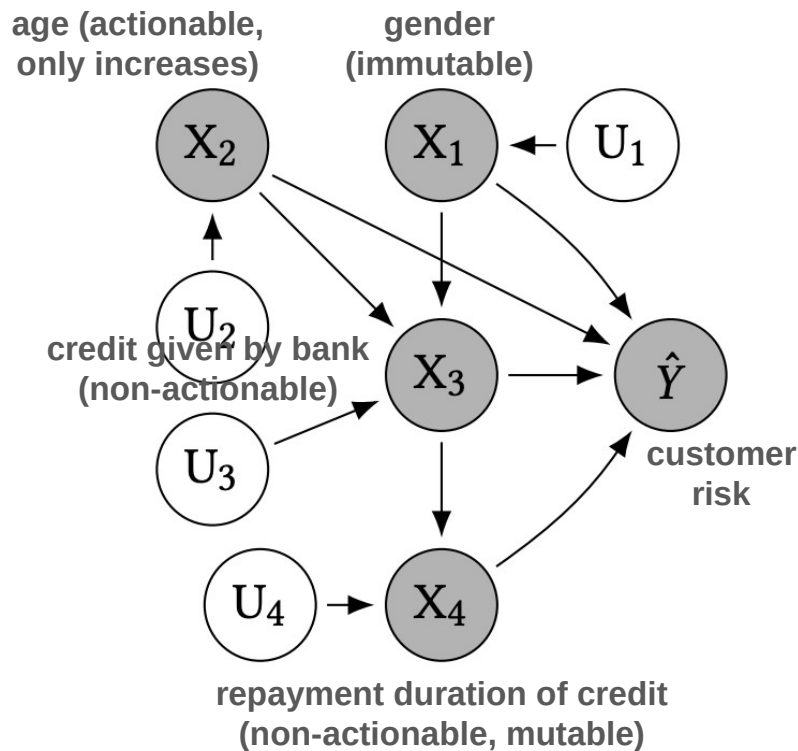
Generate data following
causal generative process

Real-world setting:

Use existing German credit
dataset to learn structural
causal model equations, by
fitting a linear regression

Cost for both is ℓ_1 norm over normalized feature change

Experimental Setup: Real-World Setting



$$\left. \begin{aligned} X_1 &:= U_1 \\ X_2 &:= U_2 \\ X_3 &:= f_3(X_1, X_2) + U_3 \\ X_4 &:= f_4(X_3) + U_4 \end{aligned} \right\} \mathcal{M}$$

$$\hat{Y} = h(\{X_i\}_{i=1}^4)$$

h can be logistic regression or decision tree

Structural Causal Model

Experimental Results: Real-World Setting

[gender, age, credit given, credit repayment duration]

$\mathbf{x}^F$		$[\text{Male}, 32, \$1938, 24]^T$
Karimi et al.'s formulation	$\mathbf{A}^*$	$\text{do}(\{X_2 := x_2^F + 1, X_3 := x_3^F - \$800\})$
	$\mathbf{x}^{*\text{SCF}}$	$[\text{Male}, 33, \$1138, 22]^T$
Prior formulation	δ^*	$[\text{N/A}, +6, \$0, 0]^T$
	$\mathbf{x}^{*\text{CFE}}$	$[\text{Male}, 38, \$1938, 24]^T$

Experimental Results: Real-World Setting

[gender, age, credit given, credit
repayment duration]

$\mathbf{x}^F$		[Male, 32, \$1938, 24] ^T
Karimi et al.'s formulation	$\mathbf{x}^{*SCF}$	[Male, 33, \$1138, 22] ^T
Prior formulation	$\mathbf{x}^{*CFE}$	[Male, 38, \$1938, 24] ^T

42% decrease in cost using
Karimi et al.'s formulation

Averaged over 50 test
individuals, $39 \pm 24\%$ and $65 \pm 8\%$
decrease in cost, for h as
logistic regression and decision
tree, respectively

Future Work: Extended Kinds of Interventions

Future Work: Extended Kinds of Interventions

Forms

- **Structural/hard** (actions in Karimi et al.): unconditionally sever all edges incident on intervened node
- **Additive/soft:** do not sever incident edges

$$x_i^{\text{SCF}} = [i \in I] \cdot \delta_i + (x_i^{\text{F}} + f_i(\mathbf{pa}_i^{\text{SCF}}) - f_i(\mathbf{pa}_i^{\text{F}})).$$

Future Work: Extended Kinds of Interventions

Forms

- **Structural/hard** (actions in Karimi et al.): unconditionally sever all edges incident on intervened node
- **Additive/soft:** do not sever incident edges

$$x_i^{\text{SCF}} = [i \in I] \cdot \delta_i + (x_i^{\text{F}} + f_i(\text{pa}_i^{\text{SCF}}) - f_i(\text{pa}_i^{\text{F}})).$$

Feasibility

Can encode as constraints to amend to $\mathbf{A} \in F$

- **Immutable:** closed under ancestral relationships; $[i \notin I] = 1$ recursively necessitates fulfillment of $[j \notin I] = 1$ for all $j \in \text{pa}_i$
- **Mutable but non-actionable:** $[i \notin I] = 1$ is sufficient
- **Actionable and mutable:** contingent on (a) pre-intervention value of variable (b) pre-intervention value of other variables (c) post-intervention value of variable (d) post-intervention value of other variables

Future Work: Extended Kinds of Interventions

Forms

- **Structural/hard** (actions in Karimi et al.): unconditionally sever all edges incident on intervened node
- **Additive/soft:** do not sever incident edges

$$x_i^{\text{SCF}} = [i \in I] \cdot \delta_i + (x_i^F + f_i(\text{pa}_i^{\text{SCF}}) - f_i(\text{pa}_i^F)).$$

Scopes

- Karimi et al. assumes action = intervention on endogenous variable
- **Fat-hand/non-atomic:** confounded/correlated interventions

Feasibility

Can encode as constraints to amend to $\mathbf{A} \in F$

- **Immutable:** closed under ancestral relationships; $[i \notin I] = 1$ recursively necessitates fulfillment of $[j \notin I] = 1$ for all $j \in \text{pa}_i$
- **Mutable but non-actionable:** $[i \notin I] = 1$ is sufficient
- **Actionable and mutable:** contingent on (a) pre-intervention value of variable (b) pre-intervention value of other variables (c) post-intervention value of variable (d) post-intervention value of other variables

Future Work: Current Limitations

Reliance on true causal
model of the world

Future Work: Current Limitations

Reliance on true causal
model of the world



True for any approach
suggesting actions to be
performed in the real world?

Future Work: Current Limitations

Reliance on true causal
model of the world



True for any approach
suggesting actions to be
performed in the real world?

Study potential inefficiencies
from partial/imperfect
causal model

Concluding Thoughts

- Overall an interesting work bridging counterfactual explanations (previous week's papers) and algorithmic recourse (next week's papers), clarifying their differences
- Tradeoff between generalizability of setting and hardness of optimization problem
 - Ustun et al. pursue algorithmic recourse in a specific linear setting
 - Karimi et al. pursue algorithmic recourse in general settings, require true causal model of world
 - Open question: where do we stand on this tradeoff? Consider the origins of counterfactual explanations in Wachter et al.